Aurélien Cécille

Computer vision researcher · self-supervised 3D perception

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PUBLICATIONS

Towards Sharper Object Boundaries in Self-Supervised Depth Estimation *BMVC 2025, Oral*

A. Cécille, S. Duffner, F. Davoine, R. Agier, T. Neveu · arXiv:2509.15987 · code

- Models per-pixel depth as a mixture distribution to remove blurred edges and spurious 3D points; up to 35% sharper boundaries on KITTI and VKITTI2.

GroCo: Ground Constraint for Metric Self-Supervised Monocular Depth *ECCV 2024*

A. Cécille, S. Duffner, F. Davoine, T. Neveu, R. Agier · arXiv:2409.14850 · code

- Metric-scale depth from a single camera without depth labels, using a ground-plane prior; improves generalisation across datasets and camera placements.

EDUCATION

Ph.D. in Computer Vision *May 2023 – June 2026*

LIRIS, INSA Lyon · industrial PhD hosted jointly with Visual Behavior

Thesis: *Structure-Aware Self-Supervised Monocular Depth Estimation*.

Master's Degree in Computer Science *2017 – 2022*

University of Technology of Troyes

EXPERIENCE

Research Engineer *Oct. 2022 – Oct. 2025*

Visual Behavior, Lyon

- Developed detection, segmentation and depth modules for mobile robotics and intelligent vehicles.
- Built local occupancy maps from depth for safe navigation around obstacles and people.
- Estimated lead-vehicle distance and detected traffic lights for energy-aware driving applications.
- Contributed to a production driver-assistance system (free-space and vulnerable road user detection).

AI Engineer Intern *Feb. 2022 – Jul. 2022*

Orange

- Detected and segmented fibre-optic cables to assess the quality of network closets.
- Evaluated few-shot and meta-learning methods; used GAN latents to improve low-annotation predictions.

AI Engineer Intern *Sept. 2020 – Feb. 2021*

Visual Behavior

- Built a monocular SLAM pipeline from self-supervised depth, pose and optical flow: camera tracking and 3D reconstruction of the scene, projected to a bird's-eye-view map.

SKILLS

Research	Self-supervised learning, monocular depth, 3D reconstruction, visual SLAM, occupancy mapping
Tools	PyTorch, JAX, Python, OpenCV, Docker, Linux, Git, OpenGL, WebGPU, D3.js
Languages	French (native), English (fluent, C2)

SUMMER SCHOOLS & WORKSHOPS

ICVSS 2025	Computer Vision for Spatial Intelligence, poster
Naver Labs 2025	AI for Robotics Workshop, poster
PAISS 2025	Attendance